HERO FINCORP

Comprehensive Data-Driven Analysis Report

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Executive Summary

This report presents a comprehensive data-driven analysis of Hero FinCorp's loan portfolio, customer behavior, branch performance, and profitability. The analysis is based on six datasets covering 90,000 loans, 70,000 customers, 50 branches, 9,000 defaults, 82,600 applications, and 495,000 transactions.

The key objective is to provide actionable recommendations to minimize loan defaults, optimize branch operations, and enhance profitability.

Key Metrics at a Glance

Total Loans	90,000
Total Customers	70,000
Total Branches	50
Total Defaults	9,000
Total Applications	82,600
Total Transactions	4,95,000
Overall Default Rate	9.49%
Overall Recovery Rate	24.31%
Total Interest Income	Rs. 7,503.75 Crores
Loan Approval Rate	84.75%
Avg Processing Time	182.8 days
Avg Days to Default	631 days

Task 1 — Data Quality and Preparation

Missing Values

The following missing values were identified across datasets:

Dataset	Column	Missing Count
Applications	Loan_ID / Approval_Date	12,600
Applications	Rejection_Reason	70,000 (for approved)
Defaults	Recovery_Status	2,985
Loans	Collateral_Details	30,194

Duplicate Records

No duplicate records were found in any dataset. All primary keys (Application_ID, Customer_ID, Loan_ID, Default_ID, Transaction_ID) are unique.

Outliers

IQR-based outlier analysis on key numeric columns showed no extreme outliers in Loan_Amount, Interest_Rate, Overdue_Amount, EMI_Amount, Default_Amount, or Recovery_Amount. This indicates the data is well-bounded and consistent.

Data Cleaning Actions Taken

- Converted all date columns to proper datetime format
- Filled missing Rejection_Reason with 'N/A' for approved applications
- Filled missing Legal_Action with 'No' for default records
- Standardized all numeric columns for analysis

Data Consistency

- Data consistency was verified across datasets using key relationships such as Customer_ID and Loan_ID.
- No major inconsistencies were found, and all records were properly aligned across tables.

Date Standardization

All date-related fields (Application_Date, Approval_Date, Disbursal_Date, Default_Date, and Transaction_Date) were standardized into date time format to ensure consistency and enable time-based analysis.

Task 2 — Descriptive Analysis

Loan Amount Distribution

Mean Loan Amount	Rs. 25,16,258
Median Loan Amount	Rs. 25,10,076
Min Loan Amount	Rs. 50,026
Max Loan Amount	Rs. 49,99,958
Standard Deviation	Rs. 14,29,288

The loan amount distribution is nearly uniform across the range, suggesting Hero FinCorp serves both small and large borrowers equally. No extreme concentration in any particular bracket.

EMI Amount Distribution

- The EMI distribution shows a wide range, indicating that customers have varying repayment capacities.
- Higher EMI values may indicate increased financial burden for certain borrowers.

Mean EMI	Rs. 50,874
Median EMI	Rs. 50,861
Min EMI	Rs. 2,001
Max EMI	Rs. 1,00,000

Credit Score Distribution

Mean Credit Score	574.56
Median Credit Score	573
Min Credit Score	300
Max Credit Score	850
Standard Deviation	159.21

The average credit score of 574 falls in the 'Fair' category, indicating a significant portion of Hero FinCorp's customers are moderate to high-risk borrowers.

Regional Loan Disbursement

Central and North regions lead in total loan disbursement. Northeast has the lowest disbursement volume, likely due to fewer branches and customers in that region.This suggests that higher loan disbursement in these regions is also associated with increased risk exposure.

Monthly Trends

Loan applications show fairly consistent monthly volumes with slight peaks observed mid-year. Approval rates remain stable at around 84-85% throughout the year. This indicates consistent lending activity without significant seasonal fluctuations.

Visualization

Histograms and bar charts were used to analyze distributions and trends, providing a clear understanding of data patterns across different variables.

Task 3 — Default Risk Analysis

Correlation with Default Flag

The following correlations were found between loan attributes and the Default_Flag:

EMI_Amount	+0.0027 (weak positive)
Overdue_Amount	-0.0007 (negligible)
Credit_Score	-0.0016 (weak negative)
Loan_Amount	-0.0033 (weak negative)
Interest_Rate	-0.0038 (weak negative)

Individual correlations are weak, suggesting defaults in this portfolio are influenced by a combination of factors rather than any single attribute.This shows that default risk is influenced by multiple factors together rather than a single variable.

Branch-Level Correlation

Delinquent Loans vs Default Rate	+0.474 (moderate positive)
Total Active Loans vs Default Rate	-0.600 (strong negative)
Loan Disbursement vs Default Rate	+0.043 (negligible)
Avg Processing Time vs Default Rate	+0.080 (weak positive)

Branches with fewer active loans tend to have higher default rates, possibly because smaller branches serve higher-risk customer segments. Delinquent loans are a strong predictor of branch-level defaults.

Pairwise Correlation Analysis

- A correlation heatmap was created to analyze relationships between key variables such as EMI_Amount, Overdue_Amount, and Default_Amount.
- The heatmap shows that while some variables have mild relationships, there are no very strong direct correlations, indicating that default behavior is influenced by multiple interacting factors.

Task 4 — Branch and Regional Performance

Regional Performance Summary

Region	Avg Default Rate	Total Disbursement	Avg Processing Time
Central	28.51% (Highest)	Rs. 3,316 Cr	9.5 days
Northeast	21.90%	Rs. 443 Cr	6.5 days
North	19.23%	Rs. 3,296 Cr	10.4 days
South	10.32%	Rs. 1,695 Cr	9.0 days
East	9.88%	Rs. 2,007 Cr	9.9 days
West	9.52% (Lowest)	Rs. 1,991 Cr	6.2 days

- Central region has the highest default rate at 28.51% despite having the highest disbursement, which suggests that customers in this region may be relatively riskier.
- West region is the most efficient with the lowest default rate and fastest processing time.

Branch-Level Performance

- Branches were further analyzed based on loan disbursement, default rates, and processing efficiency.
- It was observed that some branches with high disbursement volumes also showed higher default rates, indicating increased risk exposure.
- On the other hand, branches with lower processing times generally demonstrated better operational efficiency.

Recovery Performance

- Recovery rates vary across regions, with some regions performing better in recovering defaulted amounts.
- Regions with lower default rates generally show more stable recovery performance, indicating better credit monitoring and follow-up processes.

Task 5 — Customer Segmentation

Segmentation by Income

Low Income — Default Rate	9.60%
Lower-Mid Income — Default Rate	9.68%
Upper-Mid Income — Default Rate	9.54%
High Income — Default Rate	9.13%

Default rates are relatively uniform across income segments, ranging from 9.13% to 9.68%. High-income customers show slightly lower default rates, but the difference is not dramatic.

Segmentation by Credit Score

Very Poor (below 580)	Highest default risk
Fair (580-670)	Moderate risk
Good (670-740)	Low risk
Very Good (740-800)	Very low risk
Exceptional (800+)	Minimal risk

Segmentation by Loan Status

- Customers were also segmented based on loan status such as active, closed, and defaulted loans.
- Defaulted loan customers represent the highest risk group, while customers with successfully closed loans indicate reliable repayment behavior.

High-Risk vs High-Value Customers

High-Risk Customers (credit < 580 or defaulted)	39,676 (56.7%)
Always On Time Customers	15,297
Occasional Defaulters	7,527
Frequent Defaulters	494
Overdue But No Default	27,358

High-Value Customers

Customers with high credit scores and consistent on-time repayment behavior were identified as high-value customers. These customers are less risky and more beneficial for long-term profitability.

Overall, segmentation shows that credit score is a stronger indicator of default risk compared to income alone.

Task 6 — Advanced Statistical Analysis

An advanced pairwise correlation heatmap was generated covering Credit_Score, Loan_Amount, Interest_Rate, Overdue_Amount, EMI_Amount, Default_Amount, Recovery_Rate, and Default_Flag.

Key findings from the advanced correlation analysis:

- Credit Score has a weak negative correlation with Default_Flag, confirming lower scores slightly increase default risk
- Recovery Rate and Default_Amount are negatively correlated — higher default amounts are harder to recover
- EMI_Amount and Loan_Amount are strongly positively correlated as expected
- Interest_Rate shows minimal correlation with defaults, suggesting pricing is not a major driver

Branch-Level Correlation Analysis

- Branch performance metrics such as Delinquent_Loans, Loan_Disbursement_Amount, and Recovery_Rate were also analyzed.
- It was observed that branches with higher delinquent loans tend to have higher default rates, while loan disbursement volume shows limited direct impact on default behavior.

Task 7 — Transaction and Recovery Analysis

Total Penalty Transactions	2,47,844
Total Penalty Amount	Rs. 6,318 Crores
Recovery Rate — With Legal Action	24.48%
Recovery Rate — Without Legal Action	24.15%

Recovery Rate by Default Reason

Medical Emergency	27.76% (Highest)
Business Failure	27.31%
Other	26.54%
Job Loss	25.90% (Lowest)

Medical emergency defaults have the highest recovery rate, possibly because these borrowers intend to repay once their situation improves. Job loss defaults are the hardest to recover.

Recovery Rate by Region

South	25.86% (Best)
North	24.95%
Northeast	23.88%
Central	23.73%
West	23.32%
East	22.40% (Worst)

Branch-level recovery analysis could not be performed due to the absence of a direct relationship between loan records and branch data in the dataset.

Task 8 — EMI Analysis

EMI < 10K — Default Rate	9.70%
EMI 10K-20K — Default Rate	9.08%
EMI 20K-35K — Default Rate	9.53%
EMI 35K-50K — Default Rate	9.14%
EMI > 50K — Default Rate	9.63%

Very low EMI (below 10K) and very high EMI (above 50K) buckets show the highest default rates. This suggests both ends of the EMI spectrum carry higher risk — low EMI customers may be financially weaker, while very high EMI customers may face repayment stress.

Task 9 — Loan Application Insights

Total Applications	82,600
Approved	70,000 (84.75%)
Rejected	12,600 (15.25%)
Avg Processing Fee — Approved	Rs. 5,252
Avg Processing Fee — Rejected	Rs. 5,255

Top Rejection Reasons

Low Credit Score	4,297 (34.1%)
Incomplete Documents	4,204 (33.4%)
Insufficient Income	4,099 (32.5%)

Rejection reasons are almost evenly split between three causes, suggesting no single factor dominates. Improving documentation processes and credit awareness could reduce rejections significantly.

Task 10 — Recovery Effectiveness

Overall Recovery Rate	24.31%
With Legal Action	24.48%
Without Legal Action	24.15%

- The overall recovery rate of 24.31% is relatively low, meaning Hero FinCorp recovers only about 1 in 4 rupees from defaulted loans.
- Recovery rates for loans with legal action (24.48%) are only slightly higher than those without legal action (24.15%), suggesting that legal processes provide limited additional benefit in improving recovery outcomes.
- Recovery performance also varies across regions, indicating that operational practices and follow-up strategies at the regional level may influence recovery effectiveness.

Task 11 — Loan Disbursement Efficiency

Average Processing Time	182.8 days
Median Processing Time	184 days
Fastest Loan Purpose	Home Renovation (177.6 days)
Slowest Loan Purpose	Education (186.3 days)

- The average processing time of 182.8 days (about 6 months) is very high and represents a major delays in loan approval process.
- Education loans take the longest, possibly due to additional documentation requirements.
- The distribution of processing days also shows that while most application fall within a similar range, a few cases take longer, indicating potential inefficiencies that could be improved.

Task 12 — Profitability Analysis

Total Interest Income	Rs. 7,503.75 Crores
Most Profitable Purpose	Vehicle Loans (Rs. 1,184 Cr)
Least Profitable Purpose	Personal Loans (Rs. 1,147 Cr)
Most Profitable Region	Central Region
Least Profitable Region	Northeast Region

- Vehicle loans generate the highest interest income, making them the most profitable product,likely due to higher loan volumes and interest accumulation.
- . The Central region generates the most interest income despite having the highest default rate, indicating high disbursement volume offsets losses.
- In contrast, the Northeast region contributes the least, likely due to lower loan activity.

Task 13 — Geospatial Analysis

Active loan distribution across regions:

East	5,161 active loans
West	5,055 active loans
Northeast	4,996 active loans
North	4,988 active loans
Central	4,936 active loans
South	4,914 active loans

- Active loans are fairly evenly distributed across regions.
- However, default rates vary significantly — Central has 28.51% default rate while West has only 9.52%, highlighting regional credit quality differences that go beyond loan volume.
- This indicates that regional risk is influenced by factors beyond just loan volume, highlighting the need for region-specific credit and risk management strategies.

Task 14 — Default Trends

Monthly default counts show a gradual upward trend over the analysis period, suggesting increasing credit stress in the portfolio over time.

Average Default Amount by Loan Purpose

Education	Rs. 53,797 (Highest)
Business	Rs. 53,261
Home Renovation	Rs. 53,038
Vehicle	Rs. 52,671
Personal	Rs. 52,389 (Lowest)

- Education loans carry the highest average default amount, making them the most financially damaging defaults for Hero FinCorp.
- This suggests that while some loan categories may not default as frequently, the financial impact of defaults can still be significant, highlighting the importance of considering both default frequency and default amount in risk assessment.

Task 15 — Branch Efficiency

Total Rejected Applications	12,600 (15.25%)
Fastest Branch Processing	1 day (Sani-Kale, North)
Slowest Branch Processing	15 days (multiple branches)
Most Branches in Region	North (14 branches)

- There is significant variation in processing times across branches — from 1 day to 15 days.
- This suggests major operational inefficiencies in slower branches that need to be addressed through standardized processes and training.
- While faster processing indicates better operational speed, efficiency should also be considered along with loan quality and risk outcomes.
- Some branches may process loans quickly but could still face higher default or rejection rates.

Task 16 — Time-Series Analysis

Monthly disbursement trends show consistent lending activity with no major disruptions. Seasonal analysis reveals:

- Mid-year months (May-August) show slightly higher disbursements
- Year-end months show a minor dip in new loan applications
- Default rates across regions follow similar monthly patterns

The seasonal pattern suggests Hero FinCorp could time promotional campaigns in Q1 and Q4 to boost disbursements during typically slower periods.

Overall, understanding these time-based patterns can help improve planning, marketing strategies, and risk management decisions.

Task 17 — Customer Behavior Analysis

Always On Time	15,297 customers (30%)
Overdue But No Default	27,358 customers (53%)
Occasional Defaulters	7,527 customers (15%)
Frequent Defaulters	494 customers (1%)

Approval by Employment Status

Salaried — Approval Rate	85.0%
Self-Employed — Approval Rate	84.9%
Unemployed — Approval Rate	84.3%

- Interestingly, approval rates are similar across employment categories.
- This suggests Hero FinCorp does not significantly differentiate based on employment status, which may contribute to higher defaults among unemployed borrowers.
- Approval rates are very similar across employment categories, suggesting that employment status is not a major differentiating factor in the approval process.
- This may indicate a relatively uniform credit evaluation approach across customer types.
- Overall, customer behavior shows that a significant portion of risk lies in delayed repayments rather than outright defaults, highlighting the importance of monitoring overdue accounts closely.

Task 18 — Risk Assessment

Risk Matrix by Loan Purpose

Loan Purpose	Default Rate	Avg Default Amt	Risk Score
Education	9.95%	Rs. 53,797	0.79 (Highest Risk)
Home Renovation	10.02%	Rs. 53,038	0.69
Vehicle	10.03%	Rs. 52,671	0.42
Personal	10.11%	Rs. 52,389	0.40
Business	9.73%	Rs. 53,261	0.34 (Lowest Risk)

- The risk score combines default rate and average default amount to provide an overall measure of financial risk.
- Although personal loans have a slightly higher default rate, education loans show the highest risk score due to their higher average default amount, indicating greater financial impact when defaults occur.

Customer Risk Groups

High Risk (Credit < 580)	35,897 customers (51.3%)
Medium Risk (580-670)	11,329 customers (16.2%)
Low Risk (670-740)	8,822 customers (12.6%)
Very Low Risk (740+)	13,952 customers (19.9%)

- A significant proportion of customers fall into the high-risk category, suggesting that the portfolio has a considerable exposure to lower credit quality borrowers.
- This highlights the importance of strong credit monitoring and risk management practices.

Task 19 — Time to Default Analysis

Average Days to Default	631.4 days (~21 months)
Median Days to Default	561 days (~18.7 months)
Fastest to Default — Purpose	Personal Loans (612.9 days)
Slowest to Default — Purpose	Vehicle Loans (659.7 days)
Fastest to Default — Employment	Unemployed (620.9 days)
Slowest to Default — Employment	Self-Employed (642.3 days)

- On average, customers default about 21 months after loan disbursement. Personal loans default earliest while vehicle loans take the longest.
- This gives Hero FinCorp a roughly 18-21 month window to identify at-risk customers through early warning systems before default occurs.

Task 20 — Transaction Pattern Analysis

Customers with Irregular Repayments (>30% penalty)	43,649
Penalty Amount as % of Total Transactions	50.06%
Avg Transaction — Non-Overdue Loans	Rs. 25,462
Avg Transaction — Overdue Loans	Rs. 25,530

- The fact that 50% of total transaction value is penalty payments is a major red flag.
- It indicates widespread payment stress across the portfolio. 43,649 customers showing irregular repayment patterns suggests a systemic issue with loan affordability or customer financial health.

Recommendations

1. Reducing Loan Defaults

- Implement a credit score threshold of 620 as minimum for loan approval to reduce high-risk borrowers
- Develop an early warning system that flags accounts showing overdue fees within the first 12 months — this is the optimal intervention window before default (avg default at 21 months)
- Tighten lending criteria in the Central region which has a 28.51% default rate — consider stricter income verification and collateral requirements
- Offer financial counseling to unemployed borrowers who are approved for loans, as they have the fastest time to default
- Mandate collateral for Education loans which carry the highest average default amount of Rs. 53,797

2. Optimizing Branch Operations

- Benchmark slow branches against top performers — processing time ranges from 1 to 15 days, a 15x gap that indicates major operational inconsistencies
- Standardize loan processing workflows across all branches, especially in North region which has the slowest average processing time of 10.4 days
- Deploy dedicated recovery teams in East region which has the lowest recovery rate of 22.40%
- Increase staffing in Central region branches to handle high disbursement volumes while also improving credit quality checks
- Identify and replicate best practices from West region branches which have the best overall performance — lowest default rate (9.52%) and fastest processing (6.2 days)

3. Enhancing Profitability

- Focus marketing efforts on Vehicle and Business loans which generate the highest interest income while maintaining manageable default rates
- Introduce loyalty programs and preferential interest rates for the 15,297 'Always On Time' customers to improve retention of high-value customers
- Investigate and address the 50% penalty transaction ratio — convert penalty-heavy customers to restructured EMI plans to improve actual recovery
- Expand operations in Northeast region which has low disbursement volume (Rs. 443 Cr) but significant growth potential
- Use seasonal disbursement data to time promotional campaigns in Q4 when disbursement volumes dip, boosting year-end revenue
- Develop income-based EMI products targeting the lower income segment where default rates are slightly higher — flexible repayment schedules could reduce defaults while maintaining portfolio growth

Conclusion

This analysis of Hero FinCorp's loan portfolio reveals a financially active organization with strong disbursement volumes and interest income of Rs. 7,503 Crores. However, several areas require urgent attention:

- A 9.49% default rate with only 24.31% recovery indicates significant credit losses that need to be addressed through better underwriting
- The Central region is a major risk hotspot with 28.51% default rate despite high disbursement
- Processing times averaging 182.8 days are unacceptably high and need to be reduced to improve customer experience and competitive positioning
- 50% of transactions being penalty payments points to widespread portfolio stress

By implementing the recommendations in this report — particularly around credit risk thresholds, early warning systems, branch standardization, and customer retention strategies — Hero FinCorp can significantly reduce defaults, improve recovery rates, and enhance overall profitability.

Data-driven decision making, as demonstrated through this analysis, should be embedded into Hero FinCorp's operational culture to enable proactive rather than reactive portfolio management.